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SELECTED RECOGNITION AND CREDENTIAL

2021.10	**Young Investigator Award (Clinical Research Division)** 87th Kyushu Branch Autumn Meeting of the Japanese Respiratory Society and affiliated societies Clinical application of separation-technique analysis of recorded lung sounds to coarse crackle changes before and after dialysis
Current	**Japan Statistical Society Certificate, Grade Pre-1**

EDUCATION

2022.04-Present	**Ph.D. Program**, Anesthesiology and Intensive Care Medicine Graduate School of Biomedical Sciences, Nagasaki University
2016.04-2022.03	**M.D.**, School of Medicine, Nagasaki University

POSITIONS HELD

2026.04-Present	**Research Associate** Clinical Research Center, Nagasaki University Hospital
2024.04-Present	**Physician** Department of Anesthesiology and Intensive Care Medicine, Nagasaki University Hospital
2022.04-2024.03	**Junior Resident** Nagasaki University Hospital
Current	**Collaborative Researcher** TXP Medical Co. Ltd.

RESEARCH AREAS

Causal inference and target trial emulation; longitudinal and functional data analysis; survival analysis; clinical prediction; machine learning; large-scale electronic health record and registry studies in anesthesia and intensive care.

RESEARCH CONTRIBUTIONS

Causal inference in critical care. Led a target trial emulation comparing vasopressin initiation within 0-3 hours with initiation more than 3 and up to 6 hours after norepinephrine reached 0.25 micrograms/kg/min in MIMIC-IV and eICU-CRD. The analysis used the clone-censor-weight method and was published in Intensive Care Medicine in 2026.

Longitudinal modeling. Developed a functional data analysis based on nonlinear mixed-effects models to estimate the timeframe of lactate reduction in acute cardiovascular disease. The work was published in International Heart Journal in 2024.

Kidney disease databases. Contributed to J-CKD-DB-Ex analyses of eGFR slope prediction, eGFR discordance, serum magnesium, and proteinuria reduction in IgA nephropathy. The work includes co-first-authored reports on eGFR slope prediction, serum magnesium, and proteinuria reduction, plus a web application for individual eGFR slope estimates.

PEER-REVIEWED PUBLICATIONS

**Equal contribution. †Corresponding author. Takaya Nakashima is bolded and underlined.*

1. **<u>Nakashima T</u>†**, Ichinomiya T, Nakajima M, Shinozaki T, Shibata J, Goto T, Sato S, Hara T. Ultra-early versus early adjunctive vasopressin initiation after norepinephrine escalation in septic shock: a target trial emulation. ***Intensive Care Medicine***. Published online August 11, 2026. doi: 10.1007/s00134-026-08575-3

2. **<u>Nakashima T</u>**, Nagasu H, Hirano A, Iwagami M, Goto T, Kashihara N, Nangaku M, Hirakawa Y. The eGFR difference between creatinine-based and cystatin-C-based equations is also important in an Asian population within the J-CKD-DB-Ex. ***Scientific Reports***. 2026;16:26286. doi: 10.1038/s41598-026-57444-y

3. Nagasu H*, **<u>Nakashima T*</u>**, Ihara K, Fujimori R, Goto T, Nitta D, Kishi S, Sasaki T, Kashihara N. Prediction of estimated glomerular filtration rate slope and kidney prognosis of patients with chronic kidney disease. ***Scientific Reports***. 2026;16:8883. doi: 10.1038/s41598-026-38246-8

4. Kashihara N*, Itano S*, **<u>Nakashima T*</u>**, Goto T, Yoshihara K, Eguchi S, Iekushi K, Isaka Y, Nagasu H; J-CKD-DB collaborators. Proteinuria reduction as a surrogate endpoint for clinical study of IgA nephropathy in Japanese patients: data from the J-CKD-DB-Ex. ***Clinical and Experimental Nephrology***. 2026;30(2):309-319. Published online 2025. doi: 10.1007/s10157-025-02788-4

5. Nakahashi S, Suzuki K, **<u>Nakashima T</u>**, Hayashi Y, Tanabe Y, Tanaka A, Hashiuchi S, Yamashita C, Ito Y, Wada T, et al. A reappraisal of association between ventilator-associated events and mortality among critically ill patients using marginal structural model: multicenter observational study. ***Intensive Care Medicine***. 2025;51(10):1764-1774. doi: 10.1007/s00134-025-08074-x

6. Liu K*, **<u>Nakashima T*</u>**, Goto T, Nakamura K, Nakano H, Motoki M, Kamijo H, Matsuoka A, Ishii K, Morita Y, et al. Phenotypes of functional decline or recovery in sepsis ICU survivors: insights from a 1-year follow-up multicenter cohort analysis. ***Critical Care Medicine***. 2025;53(4):e928-e940. doi: 10.1097/CCM.0000000000006621

7. **<u>Nakashima T</u>**, Sato S, Matsui H, Mizuno A. Estimating the timeframe of lactate reduction in acute cardiovascular disease using functional data analysis based on nonlinear mixed effects models. ***International Heart Journal***. 2024;65(6):1058-1065. doi: 10.1536/ihj.23-659

8. Kishi S*, **<u>Nakashima T*</u>**, Goto T, Nagasu H, Brooks CR, Okada H, Tamura K, Nakano T, Narita I, Maruyama S, et al. Association of serum magnesium levels with renal prognosis in patients with chronic kidney disease. ***Clinical and Experimental Nephrology***. 2024;28(8):784-792. doi: 10.1007/s10157-024-02486-7

9. Kasugai D, **<u>Nakashima T</u>**, Goto T. Clinical implications of urine output-based sepsis-associated acute kidney injury. ***Intensive Care Medicine***. 2023;49(10):1263-1265. doi: 10.1007/s00134-023-07190-w

10. **<u>Nakashima T</u>**, Onagase Y, Fukahori S, Fukushima C, Iriki H, Nagae Y, Yamaguchi H, Sakamoto N, Ishimatsu Y, Yamada R, et al. Potential of a lung sound analysis system using a new recording device and signal separation as an aid to physician auscultation. [Japanese] ***Nagasaki Medical Journal***. 2022;97(1):1-6.

11. Kaneko S, Hara K, Sato S, **<u>Nakashima T</u>**, Kawazoe Y, Taguchi M, Urabe S, Nakao A, Hamada K, Yamaguchi M, Hara T. Association between preoperative toe perfusion index and maternal core temperature decrease during cesarean delivery under spinal anesthesia: a prospective cohort study. ***BMC Anesthesiology***. 2021;21:250. doi: 10.1186/s12871-021-01470-y

CONFERENCE ABSTRACT

2025.06	**Development of an eGFR slope prediction model using J-CKD-DB-Ex and visualization of risk with a web application** Nagasu H, Nakashima T, Ihara K, Fujimori R, Goto T, Nitta D, Kishi S, Sasaki T, Kashihara N. Japanese Journal of Nephrology 67(4):484. [Japanese]

BOOKS, CHAPTERS, AND ARTICLES

2026.06	**Use of generative AI in academic writing: a practical guide to accelerating study planning and literature review** Respiratory Therapy, Japanese Society of Respiratory Care Medicine. Sole author. [Japanese]
2026.05	**Reading research papers in the generative AI era: a beginner's guide for junior clinicians** Chapter in Next-Generation Orthopaedic Education. Nankodo. [Japanese]
2025.06	**Reading papers in the generative AI era** Resident Note 27(6), Yodosha. [Japanese]
2024.09	**Let the machines do the tedious work: writing papers in the Gen AI era** Igaku-Shoin. Sole-authored book. [Japanese] igaku-shoin.co.jp
2024.05	**Modern Epidemiology, 4th edition, Japanese translation** Co-translator, Chapter 18: Stratification and Standardization. Gakujutsu Tosho.
2024.09	**Let the machines do the tedious work** Igakukai Shimbun Plus series, Igaku-Shoin. [Japanese] igaku-shoin.co.jp

RESEARCH AND EDUCATIONAL OUTPUTS

Current **eGFR slope prediction web application**
Prediction and visualization tool developed from J-CKD-DB-Ex data and hosted by the Japan Kidney Association. app.j-ka.or.jp

Current **Research DX course for physicians**
Five-session on-demand course for mMEDICI covering topic development, literature search, statistical analysis, writing, and submission with generative AI tools.

2022 **Lung sound analysis system**
Recorded lung sounds were separated and analyzed to quantify coarse crackle changes before and after dialysis.

SELECTED ONGOING STUDIES

Ongoing **Clinical characteristics and outcomes of anaphylactic shock requiring intensive care**
JIPAD database study; principal investigator.

Ongoing **eGFR discordance, kidney prognosis, and longitudinal change in chronic kidney disease**
J-CKD-DB cohort study; principal investigator.

Ongoing **Multiple intraoperative acetaminophen doses and outcomes after free-flap head and neck reconstruction**
Single-center retrospective study; principal investigator.

SELECTED ORAL PRESENTATIONS

2024.03 **Causative organism and outcome in bacteremia requiring intensive care: a retrospective cohort study**
51st Annual Meeting of the Japanese Society of Intensive Care Medicine.

2023.09 **Predicting hemodynamic stabilization from lactate trajectories: functional data analysis with mixed-effects models**
Japanese Federation of Statistical Science Associations 2023.

2023.02 **A prediction model for time to cardiovascular events in cardiac sarcoidosis**
33rd Annual Meeting of the Japan Epidemiological Association.

2022.11 **Predicting recovery time in acute cardiovascular disease using functional data based on nonlinear mixed-effects models**
5th Annual Meeting of the Japanese Society for Clinical Epidemiology.

PROFESSIONAL MEMBERSHIPS

2024.04-Present **Japanese Society of Anesthesiologists**

2022.09-Present **Japan Epidemiological Association**

2021.09-Present **Japanese Society for Clinical Epidemiology**

2021.06-Present **Japanese Association for Medical AI**

2020.11-Present **Behaviormetric Society of Japan**

METHODS AND SOFTWARE

R and Python; target trial emulation; clone-censor-weight estimation; marginal structural models; survival analysis; nonlinear mixed-effects models; functional data analysis; clinical prediction and machine learning.